

Follow the instructions below:

1. The exam must be solved individually.
2. Submitting materials retrieved from the internet is plagiarism.
3. You may use all notebooks posted on the course web site and your own notebooks.
4. Solutions must be in a Jupyter notebook, with a SageMath kernel.
5. The questions are provided in a Jupyter notebook.
You may use that notebook to formulate your answers to the questions.
6. Answers must be submitted before, or at 3:40pm.
7. Submit to gradescope.
8. Not submitting any answers will by default result in a zero score.
9. During the exam no questions will be answered, so do not ask questions.

Good Luck!

1. Compute the first 10 rational approximations of $\sin(\sqrt{\pi})$ with increasing size of denominator, starting at a denominator bounded by 10 and going up to a bound of 10^{10} .

Verify the accuracy of the computed rational approximations via their relative errors.

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2. Let $p = x^3 + 8x + 9$ be a polynomial with rational coefficients.

Is p irreducible over $\mathbb{Q}$, the field of rational numbers?

Extend $\mathbb{Q}$ with sufficiently many formal roots of p so that, over this extended field, p factors into a product of linear polynomials.

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3. What is expression swell? Illustrate your definition with a good example.

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4. Draw the expression tree for $\frac{x^2 + y - 1}{x + y^2 - 1}$.

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5. Consider the expression $p = abc + 2ab + 3ac + 4bc - a^2 - b^2 - c^2$.

Without retyping p , convert p into $-c^2 + cba + 4cb + 3ca - b^2 + 2ba - a^2$.

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6. Let `s = lambda n: float(sum([exp(-(k/n)**2)*cos(3*k/n) for k in range(n)])/n)`

be a function to approximate $\int_0^1 \exp(-x^2) \cos(3x) dx$.

- (a) Define a Python function `p` which does the same as `s`. Explain why `p` is inefficient.

For correctness, compare the output of `p(1000)` and `s(1000)`.

- (b) Apply Cython to improve the efficiency of `p`. Verify the correctness.

Time the execution of the Cython function for $n = 10000$, compare with timings of `p`.

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7. Let $p = 2x^7 - 7x^6 + 4x^5 - 3x^4 - 9x^3 + 5x^2 + 8x + 1$ be an expression in x .

- (a) What is the best form to numerically evaluate p in the fastest way?

- (b) Illustrate your answer with timings of the evaluation of this form at a random number, comparing with the evaluation of p , given as a symbolic expression.

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8. Transform $p = (x + 2)^6 - y^4$ into $r = ((x + 2)^3 + y^2)((x + 2)^3 - y^2)$, without typing r .

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9. Consider the expressions $p = x + 4$ and $q = \frac{x^2 + x - 12}{x - 3}$.

- (a) Are the expressions *symbolically* the same?

- (b) Are p and q *numerically* equivalent?

Justify your answers by the appropriate computations.

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